Quartus Tutorial03A: 2-bit Adder Design GLaw
VHDL Design Entry Method with Quartus Simulation

Notation:

To simplify what one should click to perform a task, the following form of notation will be used:

Menu item → Menu item → … → Menu item

Example:

File → Project Wizard means "on the File menu, click New Project Wizard"

Objective

To learn to use Quartus to develop and simulate a structured 2-bit adder using VHDL Design Entry Method

Development Tools

Quartus 7.2

Hardware

DE2 board, USB Blaster, and USB cable.

References

Altera Quartus Handbooks Volume 1, 2, and 3 available at www.altera.com or at the course website www.csun.edu/~glaw

Launch Quartus II Software

1. Launch the Quartus II software.
2. File → Project Wizard (on the File menu, click New Project Wizard)
3. New Project Wizard: Introduction windows appears. It shows you the five steps you may need to go through in setting up a new project. You have nothing to do in this window, hence click **Next**.
4. New Project Wizard: Directory, Name, Top-Level Entity window appears.

   a. Enter the working project directory in your Z drive, for example, z:\...\Tutorial03A

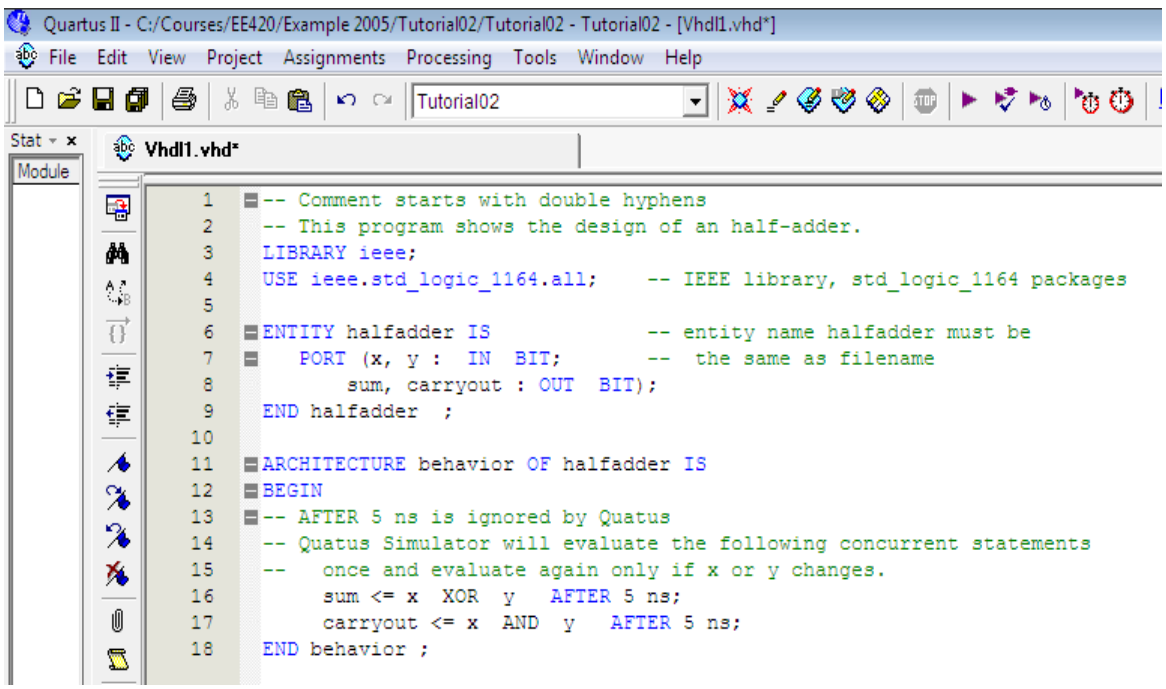
   b. Enter project name: Tutorial03A. As you type Tutorial03A name, Top-Level Entity name will be automatically filled with Tutorial03A.

   c. Click next. When asked to create z:\...\Tutorial03A directory, click **Ye**s.
5. New Project Wizard: Add Files window appears. You have no files to add now, but need to add files later. Hence, just click **Next**.

6. New Project Wizard: Family & Device Settings window appears.
   Scroll up/down the family choices and choose Cyclone II family. Scroll up/down the available devices and choose EP2C35F672C6. Leave other options as they are. Click **Finish**.

Create a Half Adder

1. File → New. New file option window appears.
2. Select VHDL File. Click OK. Quartus Text Editor window appears.
3. Starting from line 1 in the Quartus Text Editor window, type the Halfadder VHDL code example 1 in the lecture note. To save time, you could copy the contents of halfadder.vhd available on the course website and paste it in the editor window.
4. Now, your VHDL file must look similar to the one shown below:

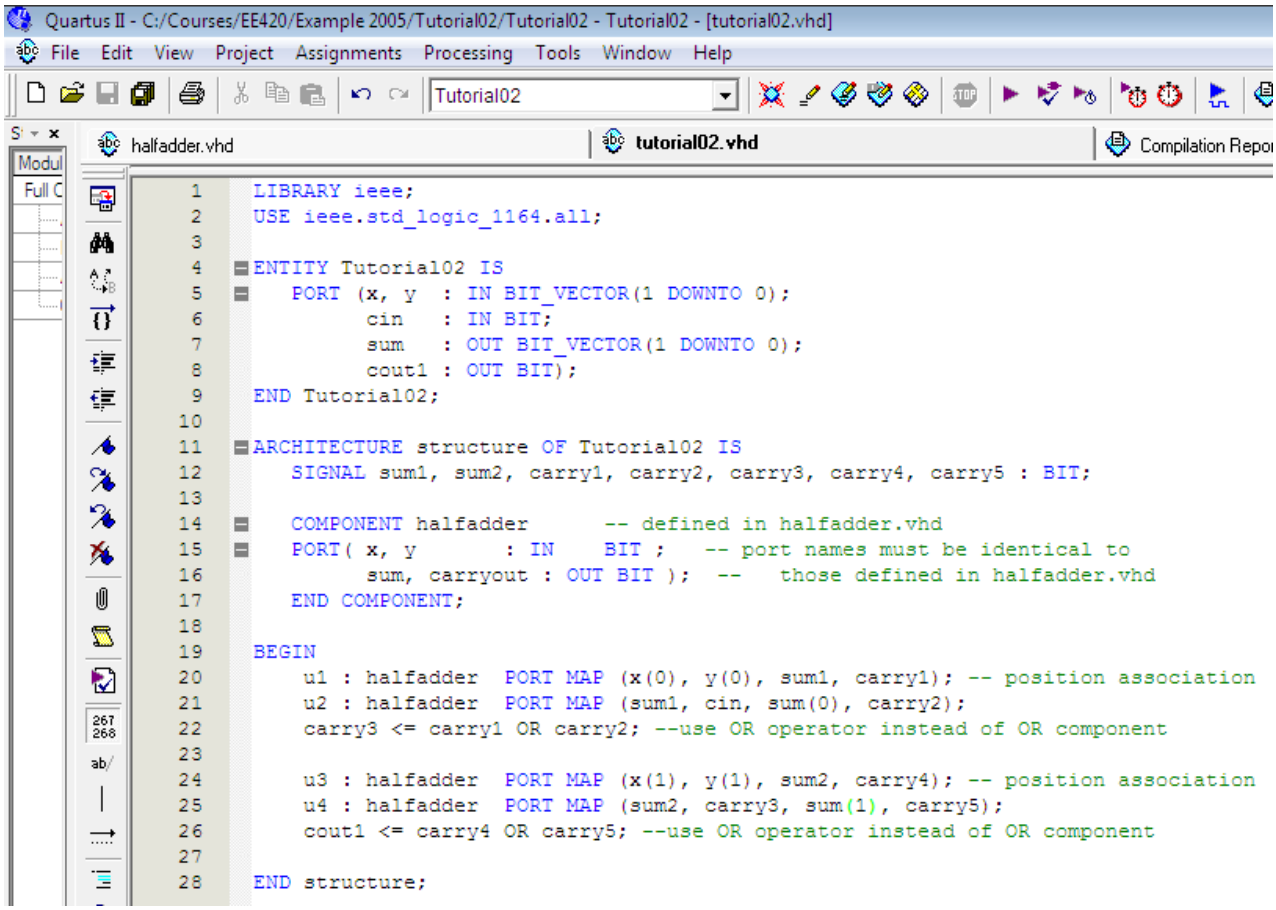


5. Save the VHDL file as halfadder.vhd.

Create a 2-bit adder using two halfadders and one OR gates.

1. File → New. New file option window appears.
2. Select VHDL File. Click OK. Quartus Text Editor window appears.

3. Starting from line 1 in the Quartus Text Editor window, type the VHDL code shown below. To save time, you could copy the contents of Tutorial03A.vhd available on the course website and paste it in the editor window. Make sure that you understand how Tutorial03A.vhd uses halfadders to implement 2-bit adder.
4. Now, your VHDL file must look similar to the one shown below:



5. Save your 2-bit adder file and name it Tutorial03A.vhd. Note that the top design name must be the same as the project name. Here, the top design is a 2-bit adder and our project name is Tutorial03A.

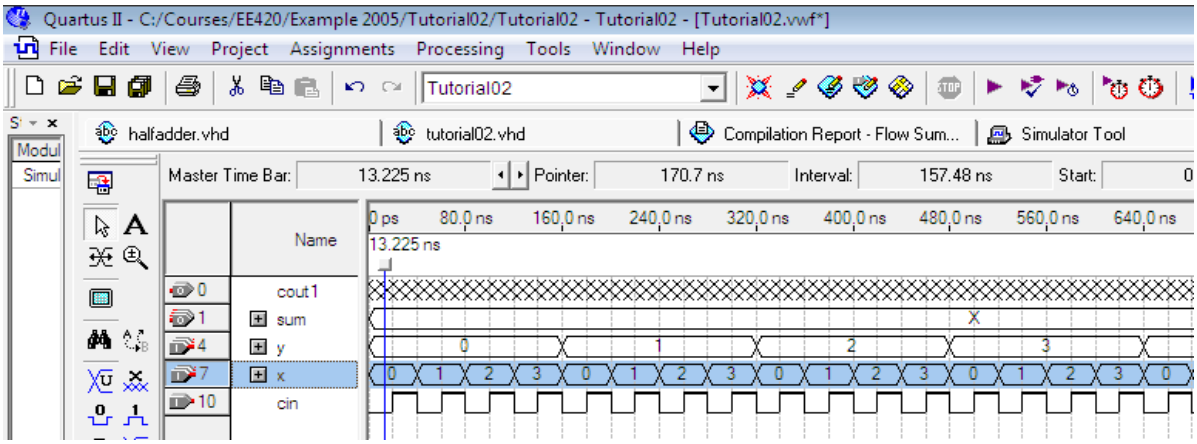
Compile the 2-bit adder

Processing → Start Compilation. If there is errors in the message window, fix the errors and compile the design again. Ignore the warnings.

Simulate the 2-bit adder

1. File → New. New file option window appears.
2. Click Other Files tab and select Waveform Vector File. Click Ok. Waveform Vector File (wvf) windows appears.
3. Left click anywhere in the Name pane (left of vwf window) and click Insert → Insert Node or Bus. Insert Node or Bus window appears.
4. Click Node Finder. Node Finder window appears.
5. Select Pins: all in the Filter selection box and click List. Select cout1, sum, x, and y signals and move these signals from the left pane (Nodes Found) to the right pane (Nodes Selected) by clicking >. Click OK. On the Insert Node or Bus window click OK again. The vwf window appears.
6. Arrange the signals in the Name pane in logical order: cout1, sum, y, x, and cin where cout1 is at the top. To move the signal up or down, select the signal, click and drag it up or down.
7. Right click on cin and Value → Count Value. Count Value window appears. Click Timing tab and change Count Every value to 20 ns. Click OK.
8. Right click on x and Value → Count Value. Count Value window appears. In the Radix box, select unsigned decimal. Click Timing tab and change the Multiplied by value to 2. Click OK.
9. Repeat step 8 for y but change the Multiplied by value to 8.
10. View → Zoom Out as many times as necessary so that you could see all possible combinations of Cin through Y in the interval 0 to 640 ns.

Your Tutorial03A vwf file must look similar to the one shown below. If there is a mismatch, go back to the previous steps to fix the errors.



11. Save your Tutorial03A.vwf file.
12. First, you will do the timing simulation where you can see the signal delay from inputs to outputs (Simulation mode: Timing). Processing → Simulator Tool. Simulator Tool window appears. Check the Overwrite simulation input file with simulation results box. Click Start. If you do not see the Start button, close the message window at the bottom.
13. If the simulation is successful, click Open to open the vwf file and see the result of simulation. If you do not see the Open button, close the message window at the bottom.
14. Check the result of simulation and take notes of the delay.
15. Second, you will do the functional simulation where all output signals change (if any) at the same time (Simulation mode: Functional). Processing → Simulator Tool. Simulator Tool window appears. Select Functional in the Simulation mode list. Click Generate Functional Simulation Netlist. After you see the message "Functional Simulation Netlist Generation was Successful", click Start to start functional compilation. If you do not see the Start button, close the message window at the bottom.
16. If the simulation is successful, click Open to open the vwf file and see the result of simulation. If you do not see the Open button, close the message window at the bottom.
17. Check the result of simulation and take notes of the zero-time delay.

Pins Assignments for DE2 Board

1. Assignments → PINs (Click PINs under Assignments)
2. Double click the blank box under Location for each of the pin names in the Table below and select the respective pin numbers. For example, to set pin number for CIN. Double click the blank box under Location for CIN node name and select PIN_N25.
3. Click OK.

Pin Assignment Table

DE2 Board Component Names	Tutorial03A Signal Names	Pin Numbers
SW[4]	Y(1)	PIN_AF14
SW[3]	Y(0)	PIN_AE14
SW[2]	X(1)	PIN_P25
SW[1]	X(0)	PIN_N26
SW[0]	CIN	PIN_N25
LEDR[2]	COUT1	PIN_AB21
LEDR[1]	SUM(1)	PIN_AF23
LEDR[0]]	SUM(0)	PIN_AE23

Fit the Project to FPGA

To launch Fitter, click Processing → Start → Start Fitter.

Assemble Project

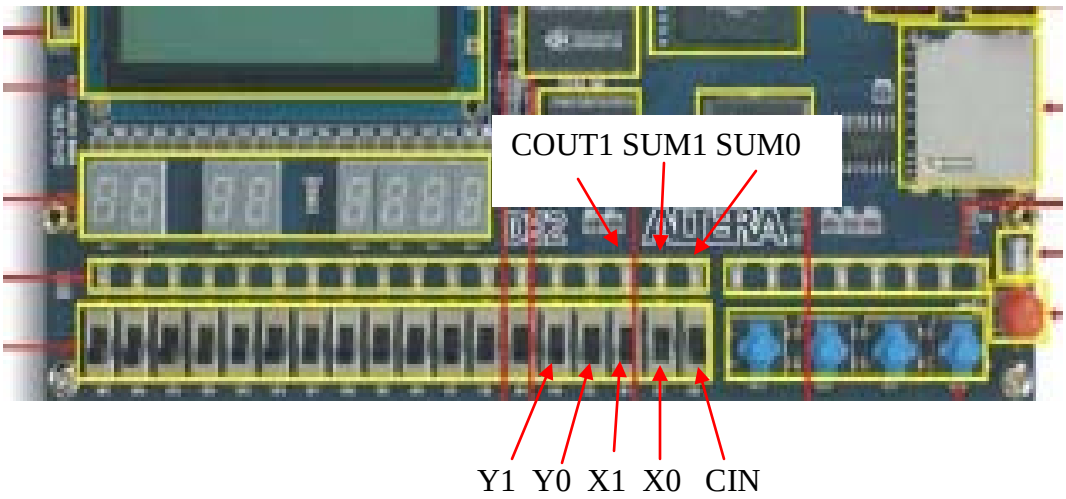
Assembler will generate the programming files (files used to configure the FPGA. For Cyclone II, it is the .sof file)

To launch Assembler, click Processing → Start → Start Assembler.

Downloading a Hardware Image to the FPGA Cyclone II 2C35F672C6

To download the Tutorial03A.sof File to the DE2 board, follow these steps:

1. On the Tools menu, click **Programmer**.
2. Click **Hardware Setup**.
3. Select **USB-Blaster** in the **Currently selected hardware** list and click **Close**.
4. Turn on (check) **Program/Configure,** located on the same row as Tutorial03A**.sof**.
5. Click **Start**.
6. Upon download success, the Messages window will display "Info: successfully performed operation(s)." If you do not see this message, check your cable connections and the Quartus II Programmer hardware setup.
7. Close the **Programmer**.

Verify 2-bit adder (Hardware verification)

Close and open SW3, SW2, SW1, and SW0 to give 16 possible combinations. For each combination, check the LEDR2, LEDR1, and LEDR0 if they match the truth table for a 2-bit adder.